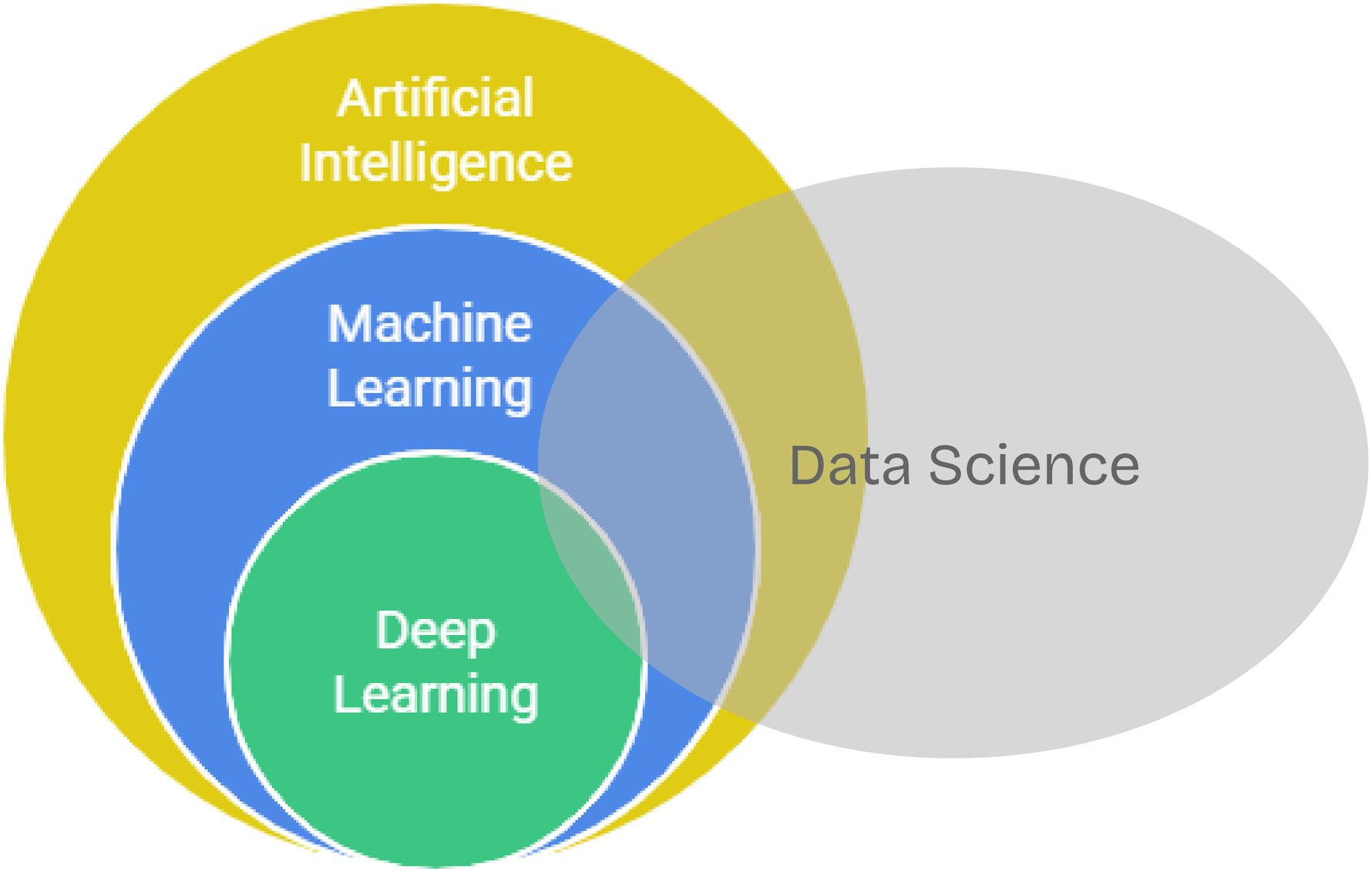
Machine Learning Unfolded

Step into the Future with AI/ML


Artificial Intelligence
Machine Learning
Deep Learning
Data Science

AI vs ML vs DL vs Data Science

Artificial Intelligence (AI) : A broader concept the "universe". AI enables an application to perform a task without human intervention.

Machine Learning (ML) : A subset of AI. It provides statistical tools to analyze, visualize, predict, and forecast the data.

Deep Learning (DL) : A subset of ML that uses multilayered neural networks that works on neural network that tries to mimic like human brain.

Data Science (DS) : A field that combines AI ML and DL to create AI-powered applications.

The main goal is to create an AI-powered application.

Types of ML Techniques

Types of Machine Learning

Supervised Learning : Uses a dataset with both independent (input) and dependent (output) features.

- **Regression :** Predicts a continuous output (e.g., house price).
- **Classification :** Predicts a categorical output (pass/fail, yes/no).
  - **Binary :** Only two output categories (Yes/No)
  - **Multiclass :** More than two output categories(Pass/Fail/May Pass)

Unsupervised Learning : Uses a dataset with only independent (input) features. There is no output/dependent feature.

- The goal is to find patterns or create clusters in the data.
- Example: Customer segmentation based on salary and spending score.

Reinforcement Learning : A technique that trains software to make decisions to achieve the most optimal results by using Trial-error method.

Example : Robots

Supervised ML Techniques

Regression :

- Linear Regression
- Ridge and Lasso Regression
- Elastic Net

Classification :

- Logistic Regression

Both Classification and Regression :

- AdaBoost
- XGBoost
- Decision Tree
- Random Forest

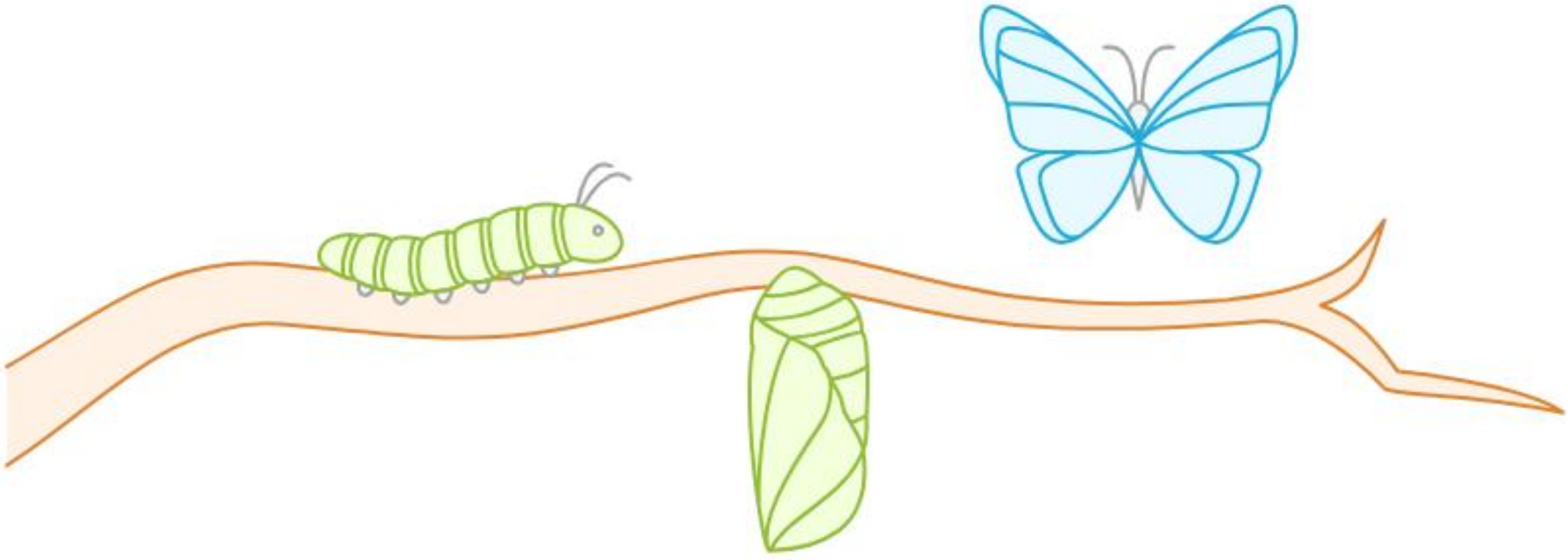
Unsupervised ML Techniques

- K-Means Clustering
- Hierarchical Clustering
- DB-Scan Clustering

Understanding ML

By using ML we make the computers learn from the data to make some predictions by which we can take any decisions easily.
Example : Predicting whether a mail is SPAM or not.
Predicting student marks based on input features.

Machine Learning Process



Input Data

Raw data for analysis

ML Model

Applying algorithms to learn

Output

Results and predictions

How our Data Looks

rollno	student_id	name	study_hours	attendance	internet_access	play_hours	assignments_completed	sleep_hours	marks
1	5008	Student_1	2	60	Yes	4	9	6	35
2	1496	Student_2	6	88	No	3	6	6	42
3	2463	Student_3	2	92	Yes	2	10	7	46
4	7828	Student_4	5	80	No	1	15	5	50
5	5253	Student_5	5	66	Yes	1	10	7	40

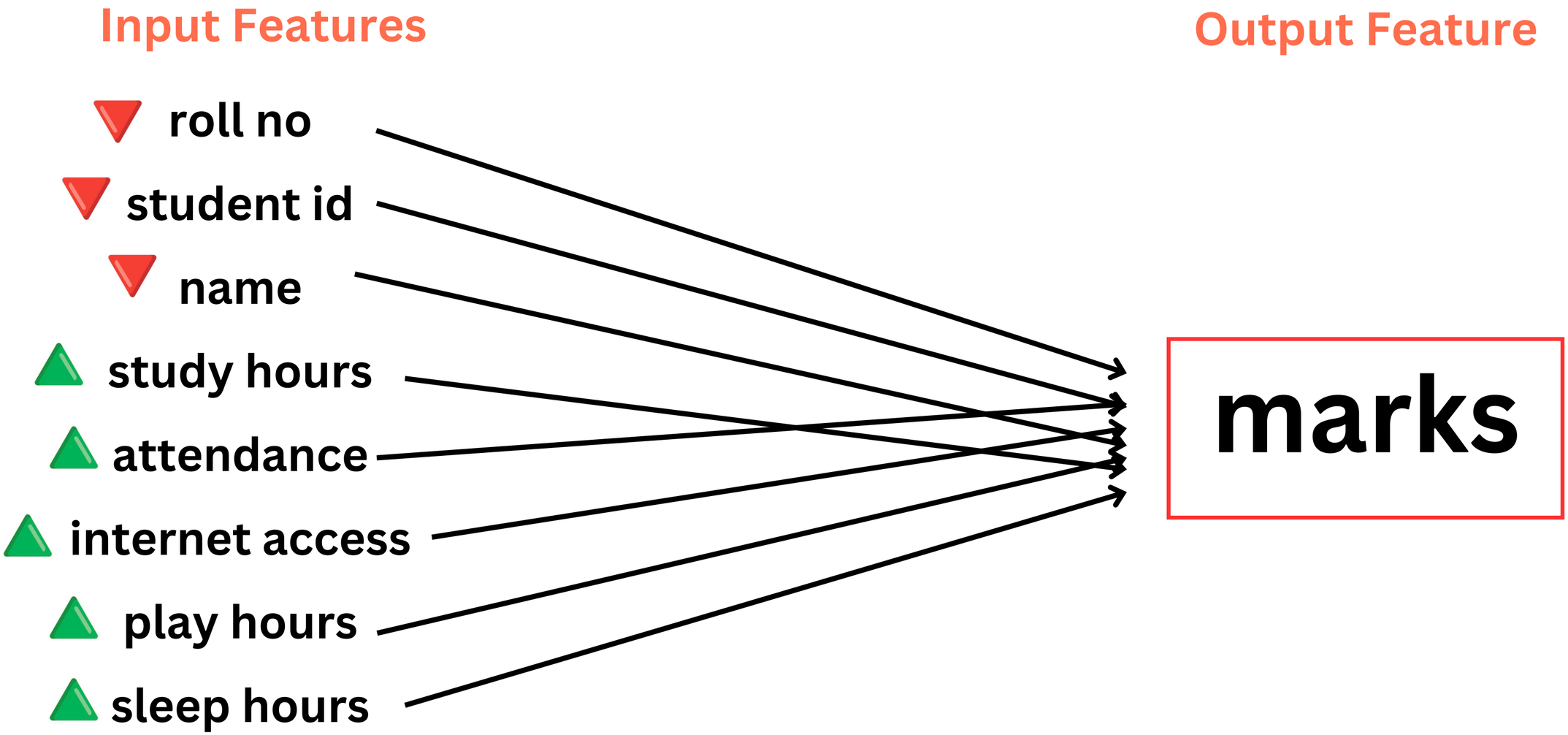
CORRELATION

Correlation is a statistical measure used to determine the relationship between two numerical variables / features .
It shows how the input feature is related with output feature.

+1 means Positive Correlation
0 means NO Correlation
-1 means Negative Correlation

Main goal is to understand how much dependent is our Output Feature with respect to Input Features.

How correlated is our I/P with O/P



Feature Selection

Feature Selection is a process in which we keep only important features (highly correlated) and remove unnecessary features.
It helps in training our model efficiently and removes noise from our data.

Unecessary columns

rollno	student_id	name	study_hours	attendance	internet_access	play_hours	assignments_completed	sleep_hours	marks
1	5008	Student_1	2	60	Yes	4	9	6	35
2	1496	Student_2	6	88	No	3	6	6	42
3	2463	Student_3	2	92	Yes	2	10	7	46
4	7828	Student_4	5	80	No	1	15	5	50
5	5253	Student_5	5	66	Yes	1	10	7	40

rollno , student_id and name have very less/No correlation with our output feature marks so we are dropping these features.

Final Data For Training

Dividing data into input and output features

Input Features(X)

Output Feature(y)

study_hours	attendance	internet_access	play_hours	assignments_completed	sleep_hours	marks
2	60	1	4	9	6	35
6	88	0	3	6	6	42
2	92	1	2	10	7	46
5	80	0	1	15	5	50
5	66	1	1	10	7	40

Training Data

Training Data is useful for training our model.

Test Data

Test Data is used to test the performance of our model , to check how good our model is trained on the training data , that helps in finding the accuracy of our model.

No. of rows in our dataset : 1000
Training dataset : 800 (80%)
Test dataset : 200 (20%)

Dividing training and test data

	study_hours	attendance	internet_access	play_hours	assignments_completed	sleep_hours	marks
Training Data 80%	2	60	1	4	9	6	35
	6	88	0	3	6	6	42
	2	92	1	2	10	7	46
	5	80	0	1	15	5	50
	5	66	1	1	10	7	40
	6	82	0	1	14	7	48
	6	81	0	4	14	7	43
	6	78	0	0	18	8	53
Test Data 20%	2	70	1	4	15	8	44
	3	70	1	3	10	7	39

Input training data
X_train

Output training data
y_train

Training Data
80%

Test Data
20%

study_hours	attendance	internet_access	play_hours	assignments_completed	sleep_hours	marks
2	60	1	4	9	6	35
6	88	0	3	6	6	42
2	92	1	2	10	7	46
5	80	0	1	15	5	50
5	66	1	1	10	7	40
6	82	0	1	14	7	48
6	81	0	4	14	7	43
6	78	0	0	18	8	53
2	70	1	4	15	8	44
3	70	1	3	10	7	39

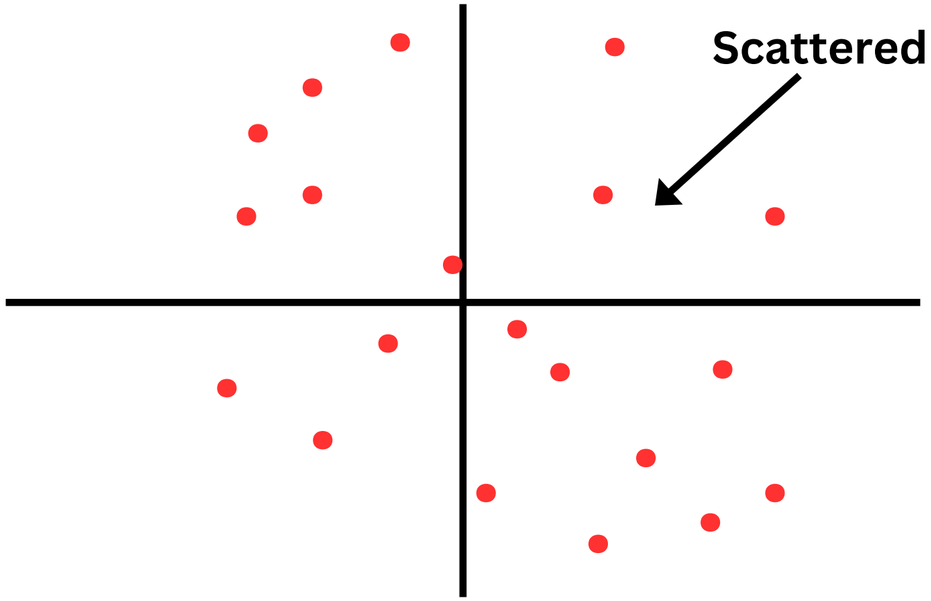
X_test
Input test data

y_test
Output test data

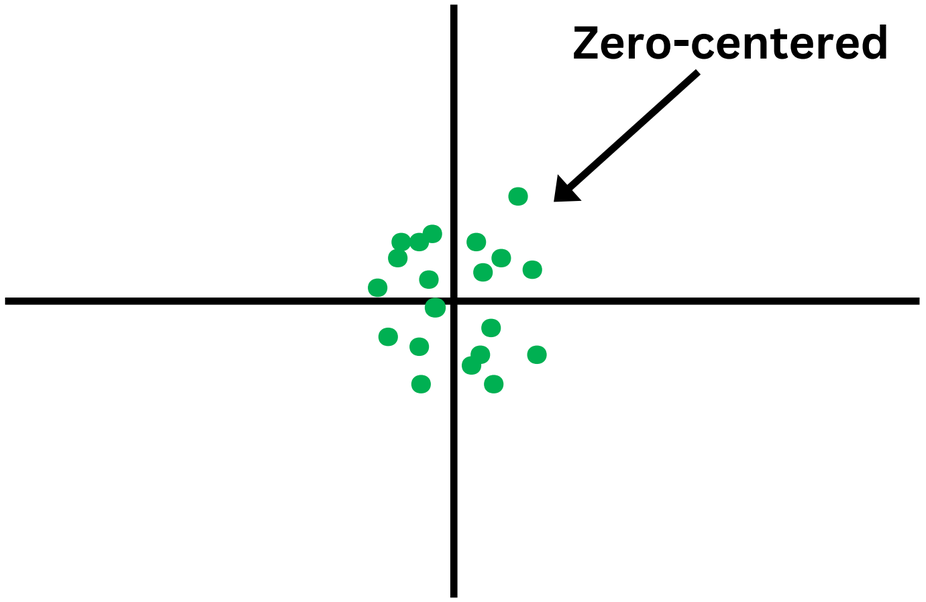
Standardization

Standardization is a technique that is applied to input features to move our data points towards center so that our model can be trained with ease and improves the accuracy of our model.

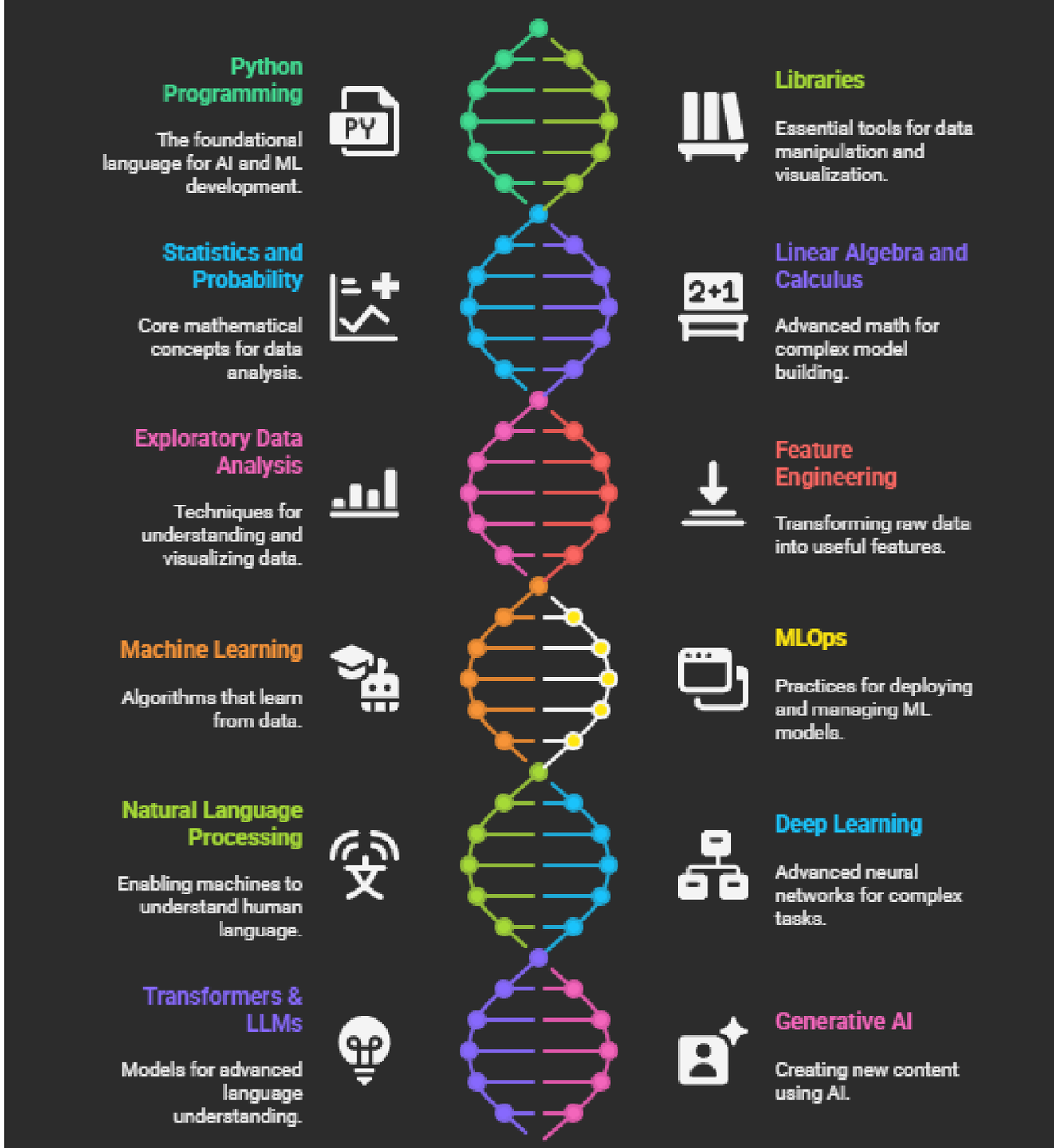
Before Standardization



After Standardization



AI & ML ROADMAP



Today's AI engineer aspirants.
GenAI
Machine Learning
Statistics
Calculus
Discrete Structure
Problem solving
Data Structure and Algorithm
python

Thank You

AI ML is the new Future